Perimeters and areas of similar figures

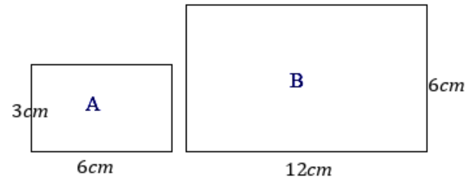
1 Consider the two similar rectangles shown.

a Find the area of Rectangle A.

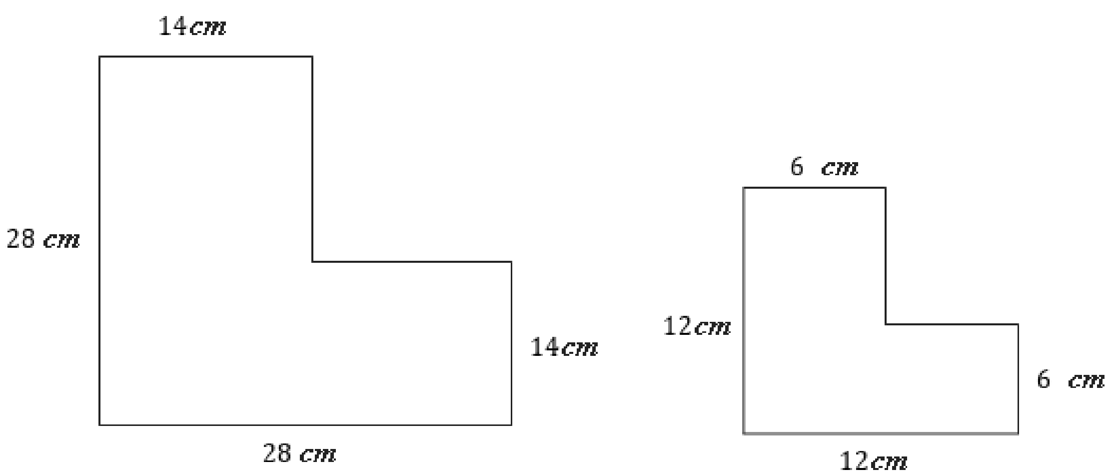
b Find the area of Rectangle B.

c What is the ratio of the area of Rectangle A to Rectangle B?

d Is it true in this case that if the matching sides of two similar figures are in the ratio $m : n$, then their areas are in the ratio $m^2 : n^2$?



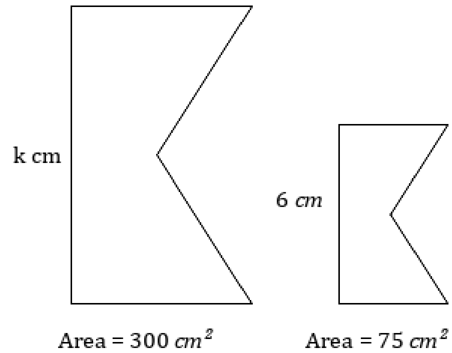
2 The corresponding sides of two similar figures are in the ratio $7 : 3$.



a State the length scale factor from the left to the right figure.

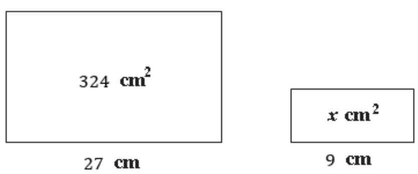
b State the area scale factor from the left to the right figure.

3 The figures in each pair are similar.
Find the value of k.

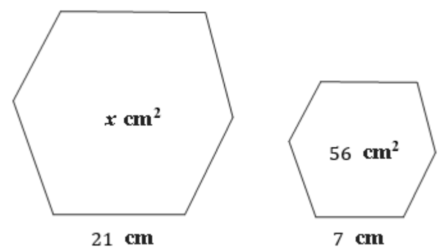


4 Find the value of x for each of the following figures:

a

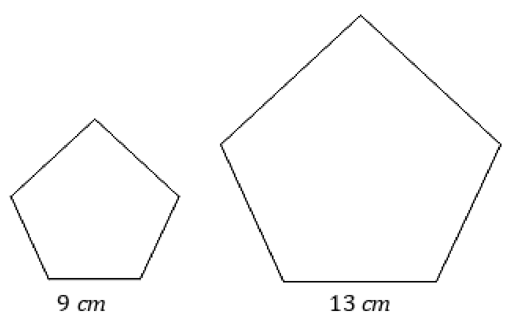


b



5 Two similar pentagons have sides of length $9\ \mathrm{cm}$ and $13\ \mathrm{cm}$.

- **a** Find the ratio between the smaller area and the larger area.
- **b** If the area of the smaller pentagon is $81\ \mathrm{cm}^2$, find the area of the larger pentagon.



6 A triangle has side lengths of $7\ \mathrm{cm}, 10\ \mathrm{cm}$ and $16\ \mathrm{cm}$. A similar triangle to it has 9 times the area of the first triangle.

- **a** Find the length scale factor between the two triangles.
- **b** Find the side lengths of the second triangle.
- **c** State the relationship between the perimeters of the two triangles

7 A square of side length $6\ \mathrm{cm}$ is enlarged using the scale factor 3. Find the area of the enlarged square.

8 Two similar parallelograms have matching sides whose lengths are in the ratio $7:2$. Find the ratio of their areas.

9 The diameter of a circle is tripled. What happens to its area?

10 Consider two similar parallelograms with a matching sides in the ratio $6:8$.

- **a** Calculate the area of the larger parallelogram, if the area of the smaller parallelogram is $72\ \mathrm{cm}^2$.
- **b** Find the length of the base of the smaller parallelogram , if the length of the base of larger parallelogram is $12\ \mathrm{cm}$.

11 The corresponding sides of two similar triangles are $8\ \mathrm{cm}$ and $40\ \mathrm{cm}$. If the area of the smaller triangle is $24\ \mathrm{cm}^2$, find the area of the larger triangle.

12 The cost of constructing a garden bed is dependent on the area it covers. A rectangular garden bed measuring $16\ \mathrm{ft}$ by $24\ \mathrm{ft}$ costs a known amount to construct. If you decided to create a similar garden bed, measuring $20\ \mathrm{ft}$ by $30\ \mathrm{ft}$, by what factor would the cost increase?

13 Consider two similar ceilings where the first has dimensions $5\ \mathrm{m}$ by $4\ \mathrm{m}$, and the second has dimensions $20\ \mathrm{m}$ by $16\ \mathrm{m}$.

- **a** Find the length scale factor.
- **b** Find the area scale factor.
- **c** If the smaller ceiling took $1.5\ \mathrm{L}$ of paint to cover it, determine the amount in liter of paint required to cover the second ceiling.

14 Consider the formula $A=\dfrac{1}{2}bh$. Determine the effect of each of the following have on A.

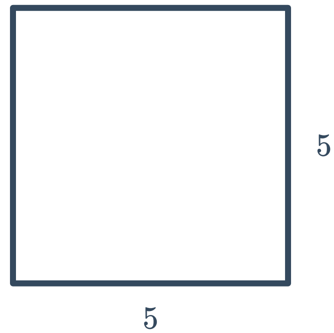
- **a** The values of b and h double.
- **b** The value of b increases by a factor of 2 and the value of h decreases by a factor of 2.

15 Consider the formula $A=\pi r^2$. Determine the effect of each of the following have on A.

- **a** The value of r triples.
- **b** The value of r decreases by a factor of 4.

Additional questions

16 Emily is knitting a rectangular blanket. So far the blanket measures $5 \text{ ft } \times 5 \text{ ft}$. If Emily increases both the length and the width of the blanket by 1 ft, how many times bigger will the area of the finished blanket be?



17 A triangle with an area of $\dfrac{3}{5} \text{ m}^2$ is dilated by a factor of 10. Find the area of the new triangle.

18 A regular pentagon has been enlarged with a scale factor of 3.25. Describe the relationship between the perimeter of the two pentagons.